

Auteur: Abdellah El Moussaoui

Studierichting: Informatica

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$$\int \frac{x^3}{\sqrt[3]{x^2+1}} dx$$

$$t = x^2 + 1 \Rightarrow dt = 2x dx \Rightarrow x dx = \frac{1}{2} dt$$

$$x^2 = t - 1$$

$$\Rightarrow \int \frac{x^2 \cdot x}{\sqrt[3]{x^2+1}} dx = \int \frac{(t-1)}{\sqrt[3]{t}} \cdot \frac{1}{2} dt$$

$$= \frac{1}{2} \int \frac{t-1}{\sqrt[3]{t}} dt$$

$$= \frac{1}{2} \int \left(t^{\frac{2}{3}} - t^{-\frac{1}{3}} \right) dt$$

$$= \frac{1}{2} \left(\frac{3}{5} t^{\frac{5}{3}} - \frac{3}{2} t^{\frac{2}{3}} \right) + C$$

$$= \frac{3}{10} t^{\frac{5}{3}} - \frac{3}{4} t^{\frac{2}{3}} + C$$

$$= \frac{3}{10} (x^2 + 1)^{\frac{5}{3}} - \frac{3}{4} (x^2 + 1)^{\frac{2}{3}} + C$$

References